

CONCEPT MAP 3: CLOUD SERVICE MODELS

Project Title:

Architectural Analysis and Deployment of Serverless Computing in Modern Cloud Platforms

CLOUD SERVICE MODELS

1. IaaS

(Infrastructure as a Service)

- Virtual Machines
- Storage
- Networking
- User manages OS & Applications

2. PaaS

(Platform as a Service)

- Runtime Environment
- Development Tools
- Middleware
- User develops applications

3. SaaS

(Software as a Service)

- Ready-to-use Software
- Web Applications
- No Installation
- Provider manages everything

4. FaaS

(Function as a Service / Serverless Computing)

- Event-Driven Functions
- Automatic Scaling
- Pay-per-Execution
- No Server Management

5. RESPONSIBILITY DISTRIBUTION

- IaaS:** Provider → Infrastructure | User → OS & Applications
PaaS: Provider → Platform | User → Applications
SaaS: Provider → Entire Software Stack
FaaS: Provider → Infrastructure & Runtime | User → Functions

6. BENEFITS OF CLOUD SERVICE MODELS

- Scalability
- Flexibility
- Cost Efficiency
- High Availability
- Faster Deployment

7. CONNECTION TO THE PROJECT

- FaaS enables Serverless Computing
- Event-driven architecture
- Automatic resource management
- Foundation for Modern Cloud Platforms

SERVERLESS COMPUTING (FaaS)

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